30.10.2026 items review

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sectionStatus of the 23.10 items

1. **Check the QUBOs against the interaction matrix. Done.** The three checks are run on the *reduced* QUBO, after qubo-solver's variable fixing, since that is what it actually embeds.
2. **Check the preprocessing. Done.** Option (b) chosen (Pass the information on to the pulse design, so that the neighbouring atoms receive the shift through local detuning).
3. **Get used to the solver. Done.**
4. ****EXTRA** Hubs. Done, not yet in the pipeline.** A hub is a user that still conflicts with more than 6 others after the fixing. Causes problems because a 2D register cannot reproduce it.

Still open from 08.10: the weight precision study (±5%) and the effect of the coverage repair on solution quality.

1 Context

1. **Checks.** When looking at the embedding, some QUBOs are set to "check", which means the comparison against the interaction matrix is different (couplings too weak/strong...), but the result is not affected, still reaches the optimum.
2. **Hubs.** In the reference run (one scenario, seed 2) hubs appear in 4 of 262 QUBOs. Whether that is rare or typical depends on how realistic the scenario generator is, so it should be checked before deciding how much effort hubs deserve.
3. **The quantum part has little work.** Most of each problem is decided classically by the preprocessing, and the QUBOs that reach the hardware have about 3–4 atoms on average. Reaching the QPU size needs larger per antenna problems.

2 Items

1. Figure out why checks happen and why those differences on the embedding don't affect the final result.
2. Figure out how often hubs appear in realistic scenarios.
3. Check whether the scenario generator is realistic; update it if not.
4. See how big the problems can go: up to about 25 qubits on the emulator and about 200 on the QPU. For every size, 10 instances (different random seeds), so results are averages with a spread rather than a single run.
5. Plug the solver into the pipeline: full assignment over all slots with qubo-solver as the oracle.

3 Benchmarking plan

Same scenario and random seed used on every solver; every schedule is checked with the same validity rules.

- Greedy oracle: the baseline.
- QUBO/MIS, solved classically (checks the QUBO formulation agrees with Greedy).
- Pulser based quantum sampling on the emulator/QPU.

Metrics: gap to the exact optimum per QUBO; total spectral efficiency, coverage (must be 100%), worst, 10th percentile and median user rate, Jain index, handovers and new users served, runtime against QUBO size.